

Foundations of Algorithmic bias

[Algorithmic bias]

1.0 Content Block

Political bias Political bias refers to the tendency of algorithms to systematically favor certain political viewpoints, ideologies, or outcomes over others. Language models may also exhibit political biases. Since the training data includes a wide range of political opinions and coverage, the models might generate responses that lean towards particular political ideologies or viewpoints, depending on the prevalence of those views in the data.

2.0 Content Block

Online hate speech In 2017 a Facebook algorithm designed to remove online hate speech was found to advantage white men over black children when assessing objectionable content, according to internal Facebook documents. The algorithm, which is a combination of computer programs and human content reviewers, was created to protect broad categories rather than specific subsets of categories. For example, posts denouncing "Muslims" would be blocked, while posts denouncing "Radical Muslims" would be allowed. An unanticipated outcome of the algorithm is to allow hate speech against black children, because they denounce the "children" subset of blacks, rather than "all blacks", whereas "all white men" would trigger a block, because whites and males are not considered subsets. Facebook was also found to allow ad purchasers to target "Jew haters" as a category of users, which the company said was an inadvertent outcome of algorithms used in assessing and categorizing data. The company's design also allowed ad buyers to block African-Americans from seeing housing ads. While algorithms are used to track and block hate speech, some were found to be 1.5 times more likely to flag information posted by Black users and 2.2 times likely to flag information as hate speech if written in African American English.

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Data bias: Training data may underrepresent certain groups (e.g., minority populations), contain historical inequalities, or be collected in skewed ways, leading the model to perform worse or behave unfairly for those groups. **Label bias:** Human-provided labels can encode subjective judgments or prejudices (for example, what is labeled as "risky," "toxic," or "qualified"), so the model learns and amplifies those judgments. **Measurement bias:** Proxies or measurements used for important concepts (like "creditworthiness" or "job performance") may be noisy or systematically distorted for some groups, which then distorts predictions. **Algorithmic bias:** Even with relatively balanced data, modeling choices (loss functions, thresholds, optimization objectives) can prioritize overall accuracy over fairness, leaving some subgroups with consistently worse outcomes. **Deployment bias:** A model used outside the context it was designed for (e.g., a model trained on adults applied to children, or one trained in one country deployed in another) can generate biased results because the environment and population differ. Mitigating machine learning bias typically involves interventions at multiple stages: collecting more representative and higher-quality data, auditing datasets and models for disparate error rates or outcomes across groups, adjusting training objectives (such as adding fairness constraints), and monitoring systems after deployment. Transparent documentation of data sources, and intended use cases is also crucial so that users and stakeholders can understand where biases may remain and how to interpret model outputs responsibly.

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Algorithmic bias describes systematic and repeatable harmful tendency in a computerized sociotechnical system to create "unfair" outcomes, such as "privileging" one category over another in ways that may or

may not be different from the intended function of the algorithm. Bias can emerge from many factors, including intentionally biased design decisions or the unintended or unanticipated use or decisions relating to the way data is coded, collected, selected or used to train the algorithm. For example, algorithmic bias has been observed in search engine results and social media platforms. This bias can have impacts ranging from privacy violations to reinforcing social biases of race, gender, sexuality, and ethnicity. The study of algorithmic bias is most concerned with algorithms that reflect "systematic and unfair" discrimination. This bias has only recently been addressed in legal frameworks, such as the European Union's General Data Protection Regulation (enforced in 2018) and the Artificial Intelligence Act (proposed in 2021 and adopted in 2024). As algorithms expand their ability to organize society, politics, institutions, and behavior, sociologists have become concerned with the ways in which unanticipated output and manipulation of data can impact the physical world. Because algorithms are often considered to be neutral and unbiased, they can inaccurately project greater authority than human expertise (in part due to the psychological phenomenon of automation bias), and in some cases, reliance on algorithms can displace human responsibility for their outcomes, without last mile thinking. Bias can enter into algorithmic systems as a result of pre-existing cultural, social, or institutional expectations; by how features and labels are chosen; because of technical limitations of their design; or by being used in unanticipated contexts or by audiences who are not considered in the software's initial design. Algorithmic bias has been cited in cases ranging from election outcomes to the spread of online hate speech. It has also arisen in criminal justice, healthcare, and hiring, compounding existing racial, socioeconomic, and gender biases. The relative inability of facial recognition technology to accurately identify darker-skinned faces has been linked to multiple wrongful arrests of black men, an issue stemming from imbalanced datasets. Problems in understanding, researching, and discovering algorithmic bias persist due to the proprietary nature of algorithms, which are typically treated as trade secrets. Even when full transparency is provided, the complexity of certain algorithms poses a barrier to understanding their functioning. Furthermore, algorithms may change, or respond to input or output in ways that cannot be anticipated or easily reproduced for analysis. In many cases, even within a single website or application, there is no single "algorithm" to examine, but a network of many interrelated programs and data inputs, even between users of the same service. A 2021 survey identified multiple forms of algorithmic bias, including historical, representation, and measurement biases, each of which can contribute to unfair outcomes.

5.0 Content Block

Surveillance Surveillance camera software may be considered inherently political because it requires algorithms to distinguish normal from abnormal behaviors, and to determine who belongs in certain locations at certain times. The ability of such algorithms to recognize faces across a racial spectrum has been shown to be limited by the racial diversity of images in its training database; if the majority of photos belong to one race or gender, the software is better at recognizing other members of that race or gender. However, even audits of these image-recognition systems are ethically fraught, and some scholars have suggested the technology's context will always have a disproportionate impact on communities whose actions are over-surveilled. For example, a 2002 analysis of software used to identify individuals in CCTV images found several examples of bias when run against criminal databases. The software was assessed as identifying men more frequently than women, older people more frequently than the young, and identified Asians, African-Americans and other races more often than whites. A 2018 study found that facial recognition software most likely accurately identified light-skinned (typically European) males, with slightly lower accuracy rates for light-skinned females. Dark-skinned males and females were significantly less likely to be accurately identified by facial recognition software. These disparities are attributed to the under-representation of darker-skinned participants in data sets used to develop this software.